

## BTEC Level 3 Applied Science • Unit 1 Chemistry • Answer sheet

## Atomic Structure — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDSscience items, not official Pearson questions.

*Numbering matches the student worksheet exactly. Award equivalent scientifically correct wording. Show units on every calculated quantity.*

**A**

1. Proton +1, mass 1. Neutron 0, mass 1. Electron –1, mass  $\approx 1/1836$ .
2. Z = number of protons. A = protons + neutrons.
3. Same proton number / element; different neutron number / mass number.

**B**

4. 19 p, 20 n, 19 e.
5. Number of protons equals number of electrons, so charges cancel.

**C**

6. Mg: 12p 12n 12e.  $\text{Mg}^{2+}$ : 12p 12n 10e. F: 9p 10n 9e.  $\text{F}^-$ : 9p 10n 10e.
7.  $A_r = (60 \times 69 + 40 \times 71) / 100 = 69.8$

|                      |                                                                  |
|----------------------|------------------------------------------------------------------|
| <b>Equation</b>      | $A_r = (\Sigma \text{ percent} \times \text{mass number}) / 100$ |
| <b>Substitution</b>  | $A_r = (60.0 \times 69 + 40.0 \times 71) / 100$                  |
| <b>Working</b>       | $(4140 + 2840) / 100 = 6980 / 100$                               |
| <b>Answer + unit</b> | 69.8 (no unit)                                                   |

Marking guidance: 1 mark substitution with both isotopes; 1 mark 69.8.

8. Chlorine is a mixture of  $^{35}\text{Cl}$  and  $^{37}\text{Cl}$ ;  $A_r$  is a weighted mean.

**D / E**

9. Isotopes must have the same proton number. C has 6 protons; N has 7.
10.  $(75 \times 63 + 25 \times 65) / 100 = 63.5$

|                      |                                                                  |
|----------------------|------------------------------------------------------------------|
| <b>Equation</b>      | $A_r = (\Sigma \text{ percent} \times \text{mass number}) / 100$ |
| <b>Substitution</b>  | $A_r = (75 \times 63 + 25 \times 65) / 100$                      |
| <b>Working</b>       | $(4725 + 1625) / 100 = 6350 / 100$                               |
| <b>Answer + unit</b> | 63.5 (no unit)                                                   |

Marking guidance: 1 mark substitution; 1 mark 63.5.

11. The atom gains two electrons. Proton number unchanged. Electrons exceed protons by 2, so charge is 2–. [3]

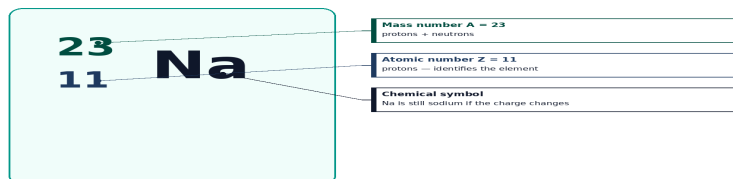
12.  $A_r = (75.0 \times 35 + 25.0 \times 37) / 100 = 35.5$ . Not whole because chlorine is a mixture of isotopes.

|                      |                                                                  |
|----------------------|------------------------------------------------------------------|
| <b>Equation</b>      | $A_r = (\Sigma \text{ percent} \times \text{mass number}) / 100$ |
| <b>Substitution</b>  | $A_r = (75.0 \times 35 + 25.0 \times 37) / 100$                  |
| <b>Working</b>       | $(2625 + 925) / 100 = 3550 / 100$                                |
| <b>Answer + unit</b> | 35.5 (3 s.f., no unit)                                           |

Marking guidance: 1 mark substitution; 1 mark 35.5; 1 mark mixture of isotopes so  $A_r$  is not an integer.

13.  $A = 23$  top left,  $Z = 11$  bottom left, 11 p, 12 n.  $\text{Na}^+$  has 10 electrons.

Nuclide notation • sodium-23



$$\text{Neutrons} = A - Z = 23 - 11 = 12$$

$$\text{Neutral atom: electrons} = Z = 11$$

$\text{Na}^+$  ion: still 11 protons; electrons = 10. The nucleus does not change.